WORKSHEET

Chapter 11: Comparing group measures

1 The heights (in cm) of two basketball teams were recorded:

Team 1	184	189	188	179	192	186	190	187	193	189
Team 2	187	185	196	189	182	191	183	185	187	194

a What is the mean height of the Team 1 players?

b What is the mean height of the Team 2 players?

c What is the standard deviation for each team?

d Compare the mean heights.

e Compare the standard deviations and state which group is the most consistent.

2 An advanced class received the following results for the end of semester exams in Maths and Science:

Maths (%)	74	85	97	92	81	100	86	93	61	78	95	83
Science (%)	81	84	87	94	76	92	91	84	52	79	91	84

a What was the mean Maths score?

b What was the mean Science score?

c What was the standard deviation for each subject?

d Compare the mean results.

e Compare the standard deviations and state which group is the most consistent.

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3 The weights (in kg) of the total tomato yield from two sets of plants fed with different fertilisers are listed below:

Set A	8.8	7.4	4.1	6.5	7.3	9.2	6.8	13.7	8.9	7.5
Set B	11.3	10.1	12.5	9.9	10.8	11.3	11.5	9.9	12.7	11.8

a What is the mean weight of the Set A tomatoes?

b What is the mean weight of the Set B tomatoes?

c What is the standard deviation for each set of tomato plants?

d Compare the mean weights.

e Compare the standard deviations and state which group is more consistent.

4 The typing speeds of a group of students, in words per minute, were measured before and after a typing course. The results were:

Original speed	32	45	29	38	37	48	51	31	26	55	41	43	36
Speed after course	38	41	38	42	32	56	62	29	38	62	50	50	47

a Draw a back-to-back stemplot.

b Does the course appear to have increased the overall typing speed of the students?

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5 The average times a group of Year 10 and Year 12 students spent on homework each night, in minutes, was recorded. The Year 10 students spent the following times:

43	97	25	62	74	79	48	92	18	47	68	34
27	63	48	66	72	58	71	60	22	51	54	71

The Year 12 students spent the following times:

49	22	98	84	91	73	68	56	94	82	76	84
18	109	95	125	72	25	93	104	129	91	86	78

a Draw a back-to-back stem and leaf plot to represent the data.

b Describe the shape of each data set.

c Which group of students spent the most time overall on homework?

d How many students spent more than 2 hours a night on homework?

e Calculate the standard deviation for each data set.

f Is Year 10 or Year 12 more consistent in the amount of time spent on homework each night?

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6 The shoe sizes of teams in two opposing basketball clubs are compared.

Club A has shoe sizes:

12 9 15 14 8 11 11 17 14 16 9 12 15
10 11 9 11 12 9 14

Club B has shoe sizes:

13 16 12 11 13 9 10 15 9 16 14 12 14
13 7 15 15 10 12 11

a Draw a back-to-back histogram to represent the data.

b Which club has the most number of players with a shoe size greater than 12?

7 A class gains the following results on a Maths test (in %):

54 63 52 74 91 47 36 59 71 59 56 95 21
42 67 61 55 51 50

a What is the median test result?

b What is Q_1?

c What is Q_3?

d Draw a boxplot to represent the data.

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It is decided an additional Maths lesson should be held each week for the class. On the next exam the students score (in %):

54 67 80 62 39 93 41 65 75 73 78 52 81
95 80 79 67 58 79

e What is the median result for this exam?

f What is Q_1?

g What is Q_3?

h On the same axes as the previous boxplot, draw a boxplot to represent the new data.

i As a result of the additional lesson, has the class improved or remained at a similar standard?

Answers

1 a 187.7 cm
b 187.9
c Team 1 = 4.1 cm, Team 2 = 4.6 cm
d On average, Team 2 is slightly taller
e The standard deviation is slightly lower for Team 1
∴ their heights are more consistent.

2 a 85.4%
b 82.9
c Maths = 11.1%, Science = 11.2%
d The class has a higher mean in Maths than Science.
e The standard deviation is slightly lower for Maths, so the group has more consistent results in Maths.

3 a 8.0 kg
b 11.2
c Set A = 2.5 kg, Set B = 1 kg
d The mean weight of Set B is higher by 3.2 kg.
e Set B has a lower standard deviation, so the plants in this group are more consistent in their yield.

4 a Key 2|6 means 26 words per minute

Original		After
9 6	2	9
8 7 6 2 1	3	2 8 8 8
8 5 3 1	4	1 2 7
5 1	5	0 0 6
	6	2 2

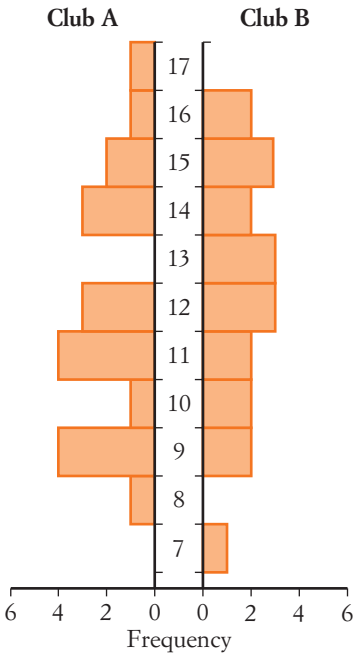
b Yes

5 a Key 1|8 means 18 minutes

Year 10		Year 12
8	1	8
7 5 2	2	2 5
4	3	
8 8 7 3	4	9
8 4 1	5	6
8 6 3 2 0	6	8
9 4 2 1 1	7	2 3 6 8
	8	2 4 4 6
7 2	9	1 1 3 4 5
		8
	10	4 9
	11	
	12	5 9

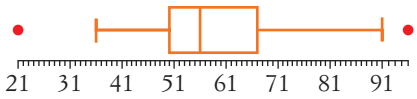
b Yr 10: symmetrical, Yr 12: negatively skewed
c Yr 12
d 2 Year 12 students
e Yr 10: 21.0, Yr 12: 28.8
f Yr 10

6 a

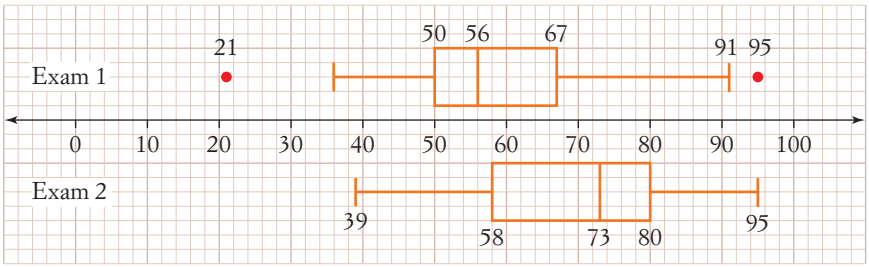


b Club B

7 a 56%
b 50%
c 67%
d



e 73%
f 58%
g 80%
h



i Improved

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